

A Friendly Primer on L^AT_EX

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How to Use L^AT_EX

You can write L^AT_EX in several ways:

- **Overleaf:** easiest option, works in your browser. Great for collaboration, reviewing, and tracking changes.
- **TeXstudio:** installable editor for offline work.
- **Git + Editor:** for advanced users, you can track your L^AT_EX projects with version control.

In this document, we will always show two things:

1. The **code you write** in `verbatim`.
2. The **compiled output** immediately after.

1 Hello World

```
\documentclass{article}
\begin{document}
Hello world!
\end{document}
```

Output: Hello world!

2 Text and Formatting

This is `\textbf{bold}`, `\textit{italic}`, and `\underline{underlined}`.

Output: This is **bold**, *italic*, and underlined.

3 Lists

```
\begin{itemize}
\item Apples
\item Bananas
\end{itemize}
```

Output:

- Apples
- Bananas

4 Math Mode

Einstein's formula: $E = mc^2$.

Output: Einstein's formula: $E = mc^2$.

```
\begin{equation}
F = ma
\end{equation}
```

Output:

$$F = ma \tag{1}$$

5 Aligned Equations

```
\begin{align}
a^2 + b^2 &= c^2 \\
e^{i\pi} + 1 &= 0
\end{align}
```

Output:

$$a^2 + b^2 = c^2 \tag{2}$$

$$e^{i\pi} + 1 = 0 \tag{3}$$

6 Figures

```
\begin{figure}[h]
\centering
\includegraphics[width=0.4\textwidth]{example.png}
\caption{An example figure.}
\end{figure}
```

Output: (requires you to have `example.png` in your folder).

7 Tables

```
\begin{tabular}{|c|c|}
\hline
A & B \\
\hline
1 & 2 \\
\hline
\end{tabular}
```

Output:

A	B
1	2

8 Code with Minted

The `minted` package lets you include highlighted code. For this you need to compile with `-shell-escape` enabled.

```
\begin{minted}{python}
import numpy as np

x = np.linspace(0, 10, 100)
y = np.sin(x)

print("Max value:", y.max())
\end{minted}
```

Output:

```
import numpy as np

x = np.linspace(0, 10, 100)
y = np.sin(x)

print("Max value:", y.max())
```

9 Citations and References

Shannon introduced information theory in 1948 `\cite{shannon1948}`.

Output: Shannon introduced information theory in 1948 [1].

References

- [1] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

Quick Cheat Sheet

- Inline math: `$...$` $\rightarrow a + b$
- Equation: `\begin{equation}...\end{equation}`
- Lists: `\begin{itemize}...\end{itemize}`
- Bold: `\textbf{...}`
- Figures: `\includegraphics{...}`
- Code: `\begin{minted}{python}...\end{minted}`